Homework 5

MATH 346/377 - Winter 2025

Due on Friday, March 28, 9:00pm on Crowdmark

Math 346: Do problems 1 to 5 – Total of points= 55 pts

Math 377: Do problems 1 to 6 – Total of points= 70 pts

1. (10 points) Find all integers $0 \leq x \leq 5^4 - 1 = 624$ such that
$$x^3 \equiv 3 \mod 5^4.$$
You can use Pari GP to help you with the computations, but you should show all the steps of your work.

2. (13 points) (a) (3 points) Show that the p-adic expansion of $\frac{1}{1-p}$ is
$$\frac{1}{1-p} = \sum_{i=0}^{\infty} p^i = 1 + p + p^2 + p^3 + p^4 + \cdots$$
   (b) (4 points) Find the p-adic expansion of $x = \frac{1+p}{1-p}$.
   (c) (6 points) Suppose that $x \in \mathbb{Z}_p$ has p-adic expansion $x = \sum_{i=0}^{\infty} x_i p^i$. What is the p-adic expansion of $-x$? In particular, what is the p-adic expansion of -1?

3. (10 points) Let $x = -\frac{2}{3}$ and $p = 5$.
   (a) (5 points) What is the 5-adic norm of x? Compute the first few terms of the 5-adic expansions of x. Based on your observations, make a conjecture about the 5-adic expansion of x.
   You can verify your solution using the Pari GP command `print(x+O(p^k))`, *which gives the first $k-1$ terms in the p-adic expansion of x.*

(b) (5 points) Prove your conjecture.

A n-th root of unity is a root ot the polynomial $x^n - 1$. In $\mathbb{C}$, all the n-th roots of unity are given by $e^{\frac{2i\pi k}{n}}$ with $0 \leq k \leq n-1$. When $p = 5$, we have seen in class that the 4-th roots unity $\pm 1, \pm i$ are contained in $\mathbb{Z}_5$. The next problem generalizes this to other primes.

4. (12 points) (a) (4 points) Use Hensel's Lemma to show that $\mathbb{Z}_p$ contains $p-1$ distinct $(p-1)$-th roots of unity μ_r, each congruent to a unique $1 \leq r \leq p-1$.

   (b) (4 points) The roots can also be described in a different way. For an integer r coprime to p, show that the sequence $x_n = r^{p^n}$ converges with respect to the p-adic norm, and that

$$\mu_r := \lim_{n \to \infty} r^{p^n}$$

   is the $(p-1)$-th root of unity that is congruent to r modulo p.

   (c) (4 points) Let $p = 7$ and $r = 2$. Use Newton's method to compute the first few terms of the 7-adic expansion of μ_2, and compare it with the 7-adic expansion of $\lim_{n\to\infty} 2^{7^n}$ given by Pari GP.
   For Newton's method you should use Pari GP to help you with the computations. You can define polynomials simply by writing `f(x)=x^6-1` *for example. On the other hand you will have a good approximation of the limit of 2^{7^n} by taking 2^{7^8}. The first five nine terms are given by the command* `print(2^(7^ 8)+O(7^10))`.

5. (10 points) Show that the polynomial

$$f(x) = (x^2 - 13)(x^2 - 17)(x^2 - 221)$$

   has a root in $\mathbb{R}$ and $\mathbb{Q}_p$ for all odd primes[1], but not in $\mathbb{Q}$.
   Hint: Quadratic reciprocity.

[1] For $p = 2$, the version of Hensel's Lemma that we have seen in class does not allow us to apply it to find roots of quadratic polynomials in $\mathbb{Q}_2$, since the derivative of $x^2 - a$ is $2x$, hence not coprime to 2. However Hensel's Lemma can be strenghtened to make it work here, and show that $x^2 - 17$ has a root in $\mathbb{Q}_2$. Thus, one can deduce that $f(x)$ has root in $\mathbb{R}$ and $\mathbb{Q}_p$ for *any* p, so it is an example of a polynomial that does not satisfy the local-global principle.

Problem 6 for MATH 377 students.

6. (15 points) Let $x \in \mathbb{Q}_p$ such that $|x|_p = 1$. The goal of this problem is to show that the p-adic expansion of x is periodic if and only if $x \in \mathbb{Q}$ and $-1 \leq x < 0$.
Remark: More generally, it is true that the p-adic expansion of $x \in \mathbb{Q}_p$ is eventually periodic *if and only if $x \in \mathbb{Q}$.*
   (a) (5 points) Show the first implication: if the p-adic expansion of x is periodic, then x is rational and $-1 \leq x < 0$.
   (b) (6 points) Show that if there is an integer $n \geq 1$ such that $(1 - p^n)x$ is an integer between and
   $$0 < (1 - p^n)x \leq p^n - 1,$$
   then the p-adic expansion of x is periodic.
   (c) (4 points) Use part b) to deduce the other direction: if $x \in \mathbb{Q}$ is such that $|x|_p = 1$ and $-1 \leq x < 0$, then x is periodic.